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Contribution	
Member	Contribution
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1. Introduction & Problem Statement

Forecasting day-ahead electricity prices is a critical challenge in the deregulated energy markets. This is due to the growing amount of variable renewable energy generation and unpredictable fuel costs over a long-time horizon [1]. Locational Marginal Price (LMP) forecasting enables market participants and stakeholders to make informed decisions that enhance the grid reliability and economic efficiency. In our first report, we demonstrated a multi-stage pipeline in which missing values for the fuel price are filled using a naïve forecast, then incorporated into a Seasonal ARIMA with exogenous regressors (SARIMA-X) framework and benchmarking the model performance against naïve and seasonal naïve models. Iterative refinements in our model scripting produced a better-in-class SARIMA-X specification that is manually added into the model—ARIMA(3,0,2)(0,1,2)[24] with external regressors, no intercept to force the model through the origin to maintain parsimony and parameters stability, faster stepwise search, and approximate likelihood—that takes 989.06 seconds to fit the autoregression model. This model forecasted the LMP with an MAE of 2.987 when compared against the known LMP values in the testing set. In terms of benchmarking, it was compared against the simple naïve model, which has an MAE of 7.358. Despite driving MAE from 4.76 to 2.987 and optimizing fitting settings, can we push the model to achieve a lower MAE without exploding model complexity or run-time? To do so, there must be a careful balance between adding information and preserving parsimony. Specifically, the next steps will systematically test small enhancements, while retaining the same model stepwise and likelihood approximation that reduced fit time to under 1,000 seconds.

2. Exploratory Data Analysis (EDA)

EDA of the three-plus years of hourly LMP and predictor series (load, solar, wind, fuel price, air temperature) revealed several key patterns and correlations that directly shaped the approach of the new model. The STL decompositions and ACF plots showed significant daily cycles and strong autocorrelations at 24-hour lags for both the LMP and load forecast. This motivated the choice of seasonal ARIMA(3,0,2)(0,1,2)[24] to capture

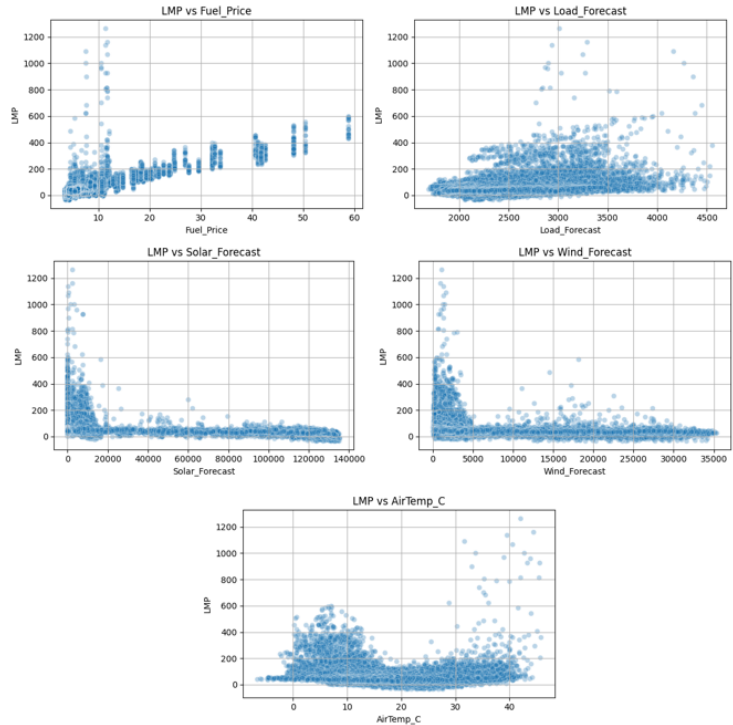


Figure 1 Scatter plots of predictors vs. LMP: (a) Top Left: scatter plot for LMP vs. Fuel Price, (b) Top Right: scatter plot for LMP vs. Load Forecast, (c) Middle Left: scatter plot for LMP vs. Solar Forecast, (d) Middle Right: scatter plot for LMP vs. Wind Forecast, (e) Bottom: scatter plot for LMP vs. AirTemp_C

intraday resolution. In addition, scatter plots, see figure 1, highlighted that load forecasts and fuel price have the strongest positive linear relationships with LMP, while solar and wind forecasts are inversely related, and air temperature exhibits a complex relationship with LMP. This motivated us to retain all five regressors to fit the model. Finally, visual inspection of weekdays vs. weekend patterns in the STL decomposition plots suggested systematic changes in price dynamics. This lead us to engineer an “Is_Weekend” dummy variable during preprocessing to be captured in the model fitting. These EDA-driven decisions ensure that the SARIMA-X model captures both the seasonality and impacts of each exogenous variable.

3. Models & Methodology

Data analysis motivated the implementation of a linear time series regression with SARIMA errors (SARIMA-X) model to forecast the LMP with a naïve forecast for fuel price. Once again, the primary factors in model selection are the strong seasonality and presence of insightful covariates. The primary covariate of interest is fuel price since, as demonstrated earlier, it has the highest correlation to LMP. The relatively short-term stability of the fuel price and constant behavior over a day motivates the use of the naïve forecast. Over the short testing window, this has proved to be an effective approach as the price has shown minimal variability, staying within a range of \$0.40 over the 3-day testing window that has been observed. Before training forecasting models, the selection of a training data window is chosen to be from January 1st, 2023, onwards to the start of the day ahead prediction. A survey of other electricity price forecasts reveals that statistical models can have smaller training periods for day ahead forecasts as older data becomes less relevant. This decision is backed up during model training as the two-year mark appears to be the sweet spot for balancing the quantity of data and relevance to the present. The reduced training window also allows for increased computational speed, allowing for more time to be spent fine tuning the model parameters. For the selection of a primary forecasting model, the only other model outside of the SARMIA-X model that can capture both seasonality, and the covariates is the harmonic regression model or the dynamic harmonic regression model. However, after many attempts to fit the harmonic regression model to the dataset, residual analysis revealed that it was not capturing the seasonality well, despite the inclusion of trigonometric terms meant to capture the seasonality. Even attempts to include lagged regressors prove to be futile, and the fit of the model is poor. The ACF plot of the

residuals to the right shows the strong seasonality and poor fit from the harmonic regression model still present, as noted by significant lags at every timestep with increasing autocorrelation in step with the 24-hour seasonal period. With the harmonic regression model not viable, the primary candidate that remains is the chosen

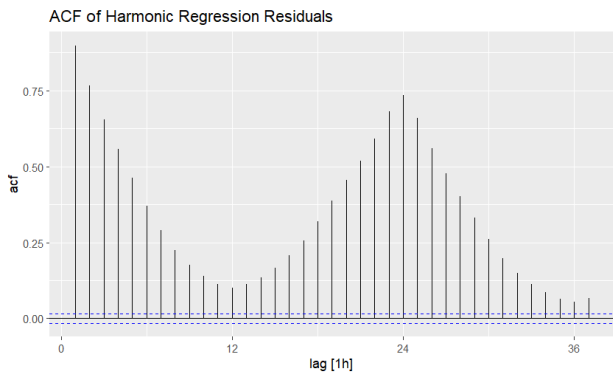


Figure 2 Residual ACF plot for a fitted Harmonic Regression model

SARIMA-X model. As an aside, models that do not account for regressors like the SARIMA demonstrated similar poor performance after brief testing, so the focus is placed on the SARMIA-X. Ultimately, after several rounds of experimentation, a SARIMA(3,0,2)(0,1,2)[24] is chosen as the primary forecasting model. The following sections will dive deeper into the experiments that support this decision, but from a scientific perspective, there are a few reasons this model is chosen. First, the obvious daily seasonality brought about by environmental factors leads to the seasonal period of 24 hours being chosen. Then, a seasonal difference is applied to stabilize the variance, which is known to improve model performance. Although the LMP series is relatively stationary, it demonstrates some volatility, and seasonal differencing largely stabilizes this. Figure 3 shows the comparison between the LMP with and without seasonal differencing:

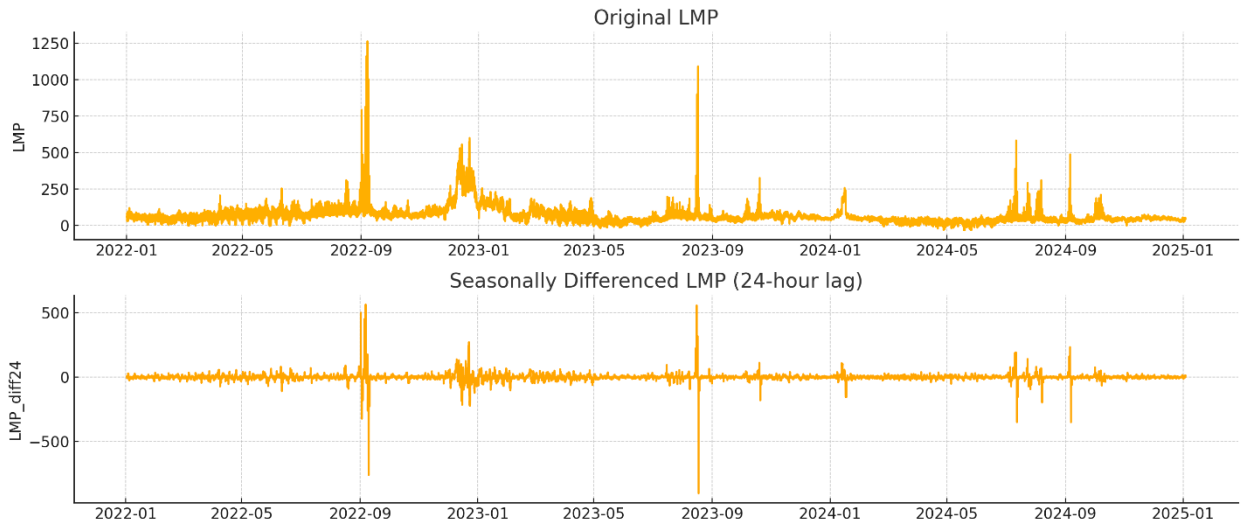


Figure 3 A comparison of timeseries plots between (a) Top: the raw LMP data and (b) Bottom: the seasonal differenced LMP data

Although some spikes still remain, they are less prevalent and smaller in magnitude in the difference data vs the non-differenced data. Another key aspect of the SARIMA-X model selection is the inclusion of three autoregressive terms. Figure 4 shows the results of the partial autocorrelation function (PACF) analysis of the LMP training data. It can clearly be seen that the third lag is the last significant lag for LMP, outside of the lags later in the period corresponding to seasonality. As a result, a p of three is chosen for the number of autoregressive terms in the model. Finally, the moving

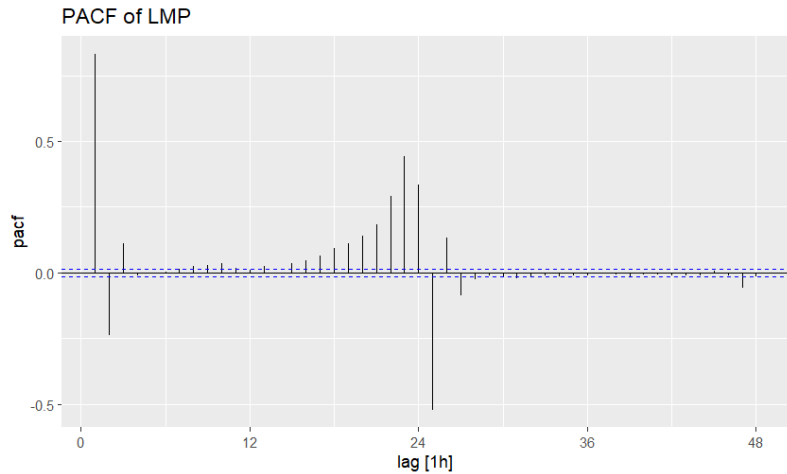


Figure 4 The PACF plot for the LMP training data

average terms, which capture the short-term dependencies in forecast errors, are chosen based on experimentation and analysis of the residuals, to be discussed in the following section. As for the regressors, all five that are provided (Solar and Wind Forecast, Air Temperature, Fuel Forecast, and Load Forecast) are incorporated into the model as they all demonstrate at least modest cross-correlation with LMP and help capture features like seasonality.

4. Validation & Experiments

To validate the model, two primary tools are used to evaluate model performance. First, a residual analysis is conducted that investigates the time series plots, ACF plots, and histograms of the residuals. A good model shows residuals that resemble white noise (i.e. no clear signal present in the data). The second is a rolling ex-post forecast that evaluates the MAE of the model over the three-day testing window that is already known. Diving into the experiments, once data analysis sets the stage for the baseline model, experiments are performed to fine-tune the model. Many different combinations of model varieties are tried, ranging from the inclusion of lagged regressors and Fourier regressors to differencing of the regressors and transformations to the LMP. Figure 5 shows the ACF plot of the residuals from the chosen model:

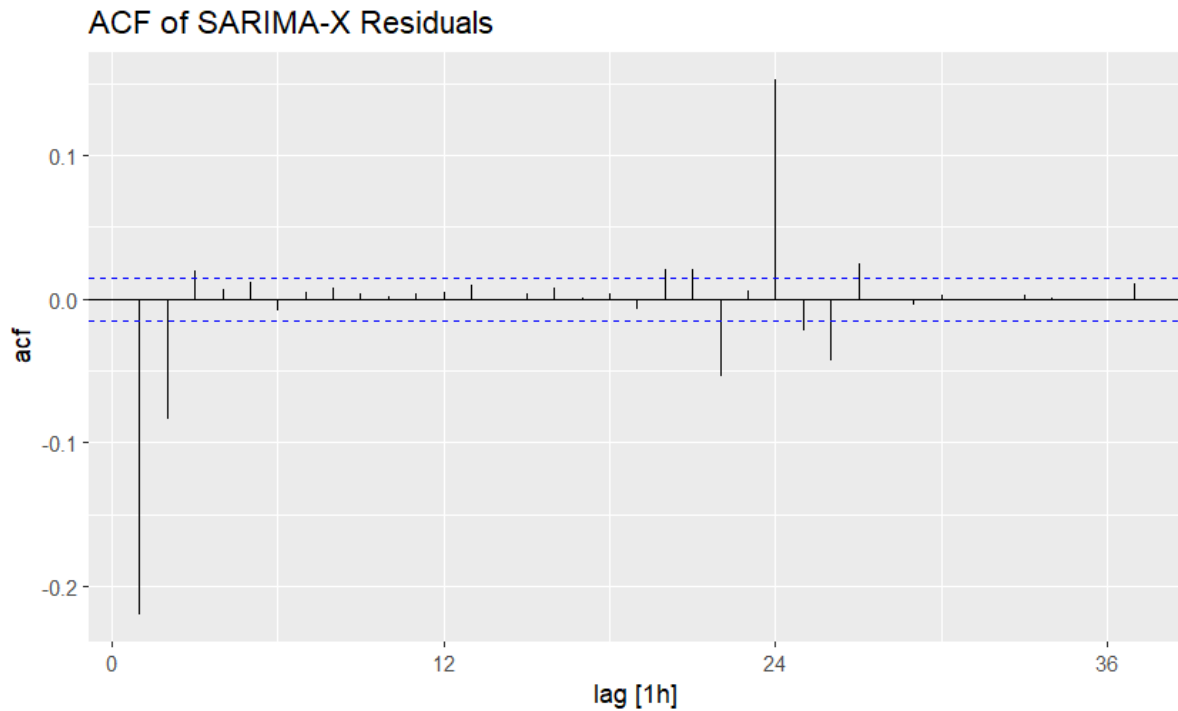


Figure 5 The ACF plot of the residuals from the chosen SARIMA-X model

One can observe that nearly all of the signals are captured by the model outside of a couple of lags at the start and at the 24-hour mark, showing adequate performance. Efforts to better capture the seasonality like introducing lagged variables, increasing the seasonal moving average or introducing Fourier terms, typically resulted in worse model performance across other lags

that offset the benefits from improved seasonal recognition. Likewise, using transformed regressors like seasonal differences on solar and wind forecasts resulted in decreased model performance on the residual plots, leading to the model that was ultimately chosen.

Ex-post forecasting is performed where the forecasts can be assumed to be known since the data is available for 3 days of testing that have already passed. Table 1 shows the results of the ex-post forecast of the implemented SARIMA-X model on the previous 3 days of test forecasting:

Forecast Week	MAE	MAE Naive
1	4.685	5.992
2	2.487	2.731
3	4.193	6.757
Average	3.78	5.10

Table 1 SARIMA-X Ex-post Forecasting Results compared to the Naïve Forecast

Table 1 demonstrates strong performance for the SARIMA-X model, as it consistently outperforms the naive benchmark model, further justifying its robustness for forecasting the LMP.

5. Conclusion

This project showed how a data-driven and EDA-informed modeling approach can be both powerful and practical enough to forecast day-ahead LMP. Beginning with over three years of hourly price and exogenous data, we narrowed down the training time range to include data points from 2023 onwards. We revealed clear 24-hour seasonality, strong positive correlations between load forecasts and fuel prices, inverse effects from solar and wind covariates, and meaningful weekend shifts. These insights guided the decision of including the five regressors in the model plus a dummy indicator variable for weekends. Through iterative comparisons, we found that harmonic regression models struggled to capture the dominant intraday cycles, whereas the $ARIMA(3,0,2)(0,1,2)$ [24] model with no intercept balanced complexity and interpretability. Residual diagnostics revealed near-white-noise behavior, which confirms that nearly all systematic dynamics were captured by the model. A rolling ex-post forecasting over a three-day period validated the model's performance. By using a naïve forecast for the fuel prices rather than more complex preprocessing, the model framework maintained simplicity without sacrificing forecasting performance. Ultimately, this project underlines the value of using both exploratory analysis with classical time series frameworks to execute interpretable and actionable forecasts. For future extensions and enhancements to the autoregressive model, hybridizing the SARIMA-X model with other non-linear capturing models could further improve the LMP forecasts and achieve a lower MAE.

References

- [1] Weron, R. (2014). *Electricity price forecasting: A review of the state-of-the-art with a look into the future*. International Journal of Forecasting, 30(4), 1030–1081.
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